Lecture 8

8.1 Discrete Signal 離散信号

*8.2 Sampling Theorem サンプリング定理

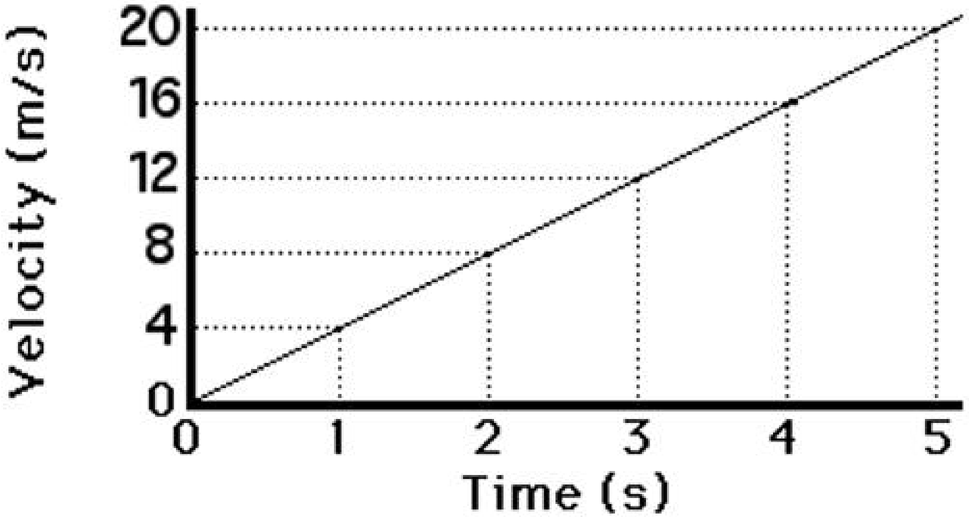
8.3 Discrete Fourier Transform 離散フーリエ変換

8.1 Discrete Signal

離散信号

8.1 Discrete Signal 離散信号

The physical world is replete with signals, that is, **physical quantities** (dependent variable (従属変数)) that change as a function of time, space, or some other **independent variable (独立変数)**.



Dependent Variable y (i.e. $f(t)$)

Independent Variable t

8.1 Discrete Signal 離散信号

- Signals whose dependent and independent variables are *continuous* are usually referred to as continuous-time signals (連続時間信号), and we will denote as $f(t),\ t \in \mathbf{R}$. Here $\mathbf{R}$ is the set of real numbers.
- In contrast, **signals where both the dependent and the independent variables are *discrete*** are called **digital signals.**
- If **only the independent variables are specified to be *discrete***, then we have a **discrete signal (離散信号).**

8.1 Discrete Signal 離散信号

- The scalar discrete signal in which **the independent variable is time** refer to as **discrete-time signal (離散時間信号)** $f(n)$, $n \in \mathbf{Z}$. Here $\mathbf{Z}$ is the set of integer numbers (整数).

Such signals usually arise when we **sample continuous-time signals**, that is, when we **select values at discrete-time instances.**

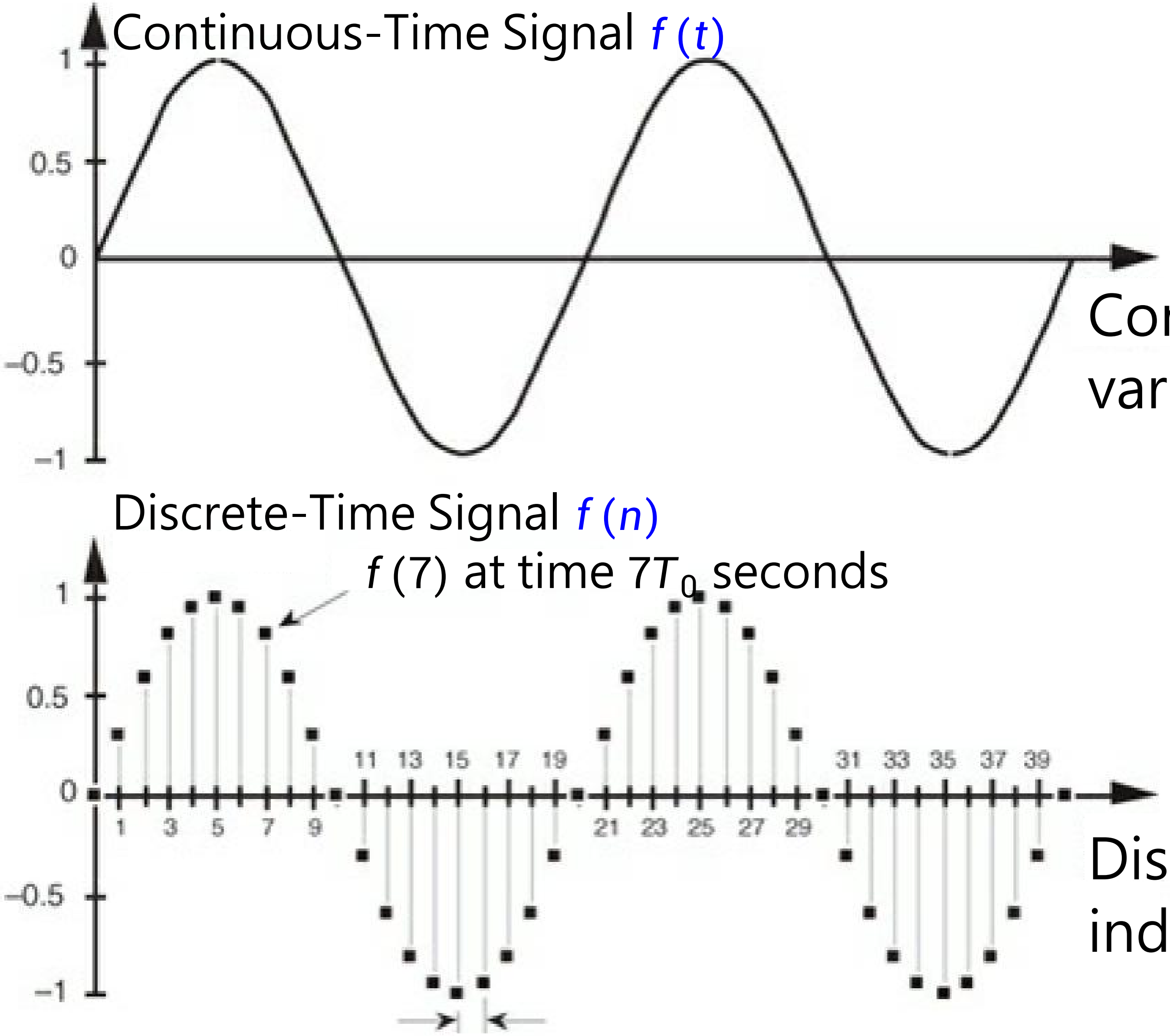


Figure 8.1

T_0 is constant time interval between samples.

8.1 Discrete Signal 離散信号

Review *Calculus II*

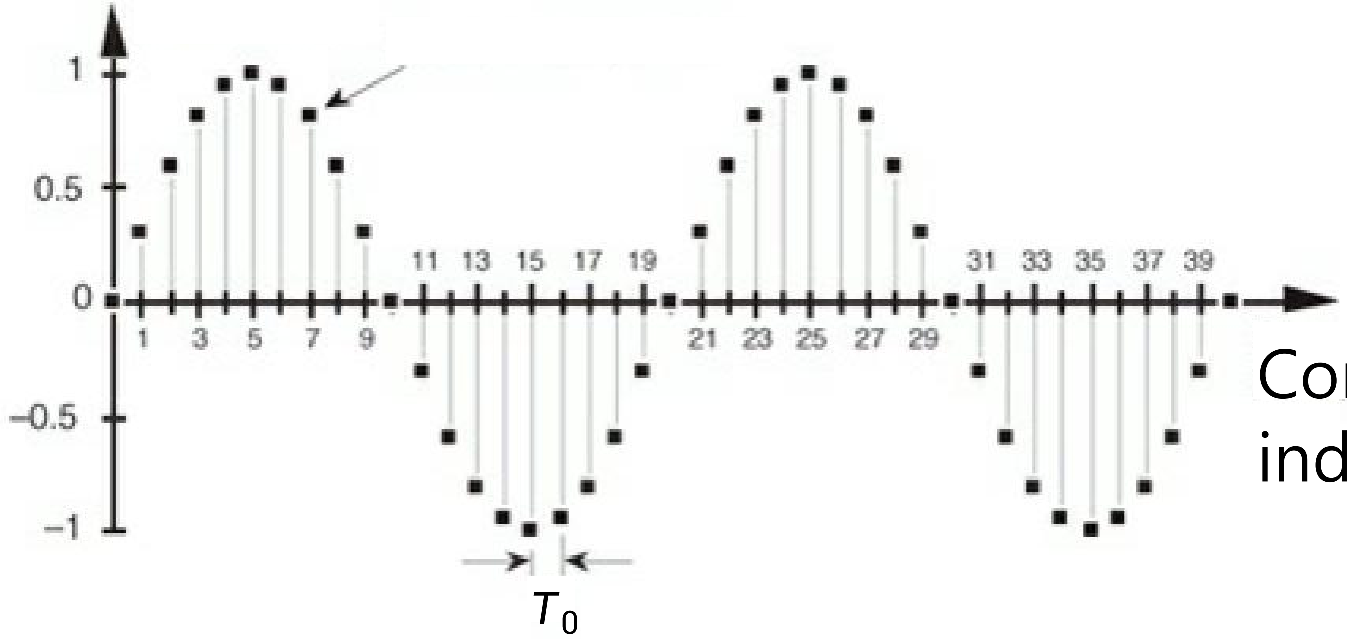
f (*n*) is a **discrete-time sequence (離散時間シーケンス)** of individual values, and **each value in that sequence plots as a single dot.**

A **sequence:** a list of numbers written in a definite order

$$\{\underbrace{a_1}_{\text{The first term}}, \underbrace{a_2}_{\text{The second term}}, a_3, a_4, \dots, \underbrace{a_n}_{\text{The } n\text{th term}}, \dots\}$$

Notice that each term a_n will have a successor a_{n+1}.

Discrete *f* (*n*)



Continuous-time index, *n*

8.1 Discrete Signal 離散信号

Discrete Signal in Real-World

8.1 Discrete Signal 離散信号

Periodic Functions **Review Lecture 1**

In general, a function f **satisfying the identity**

$$\sin x = \sin(x + T) \quad \text{for all } x \qquad (1.1)$$

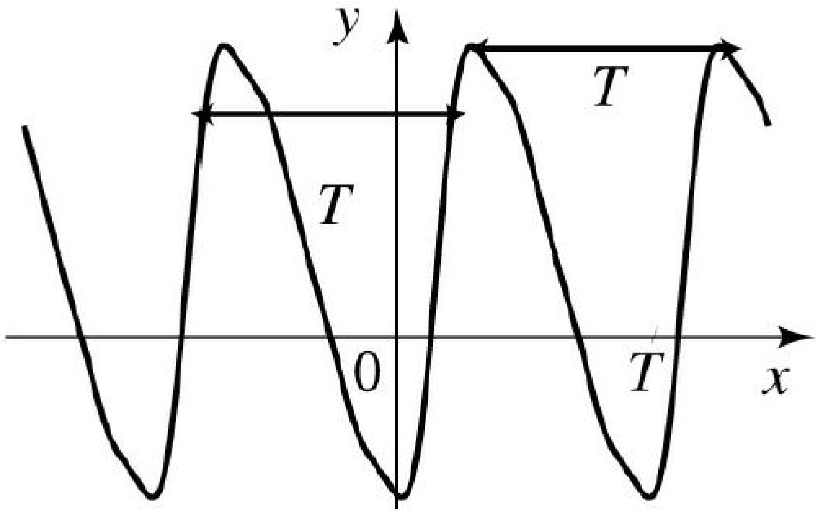


Figure 1.2 T-periodic function

where $T > 0$, **is called periodic**, or more specifically, T-periodic.

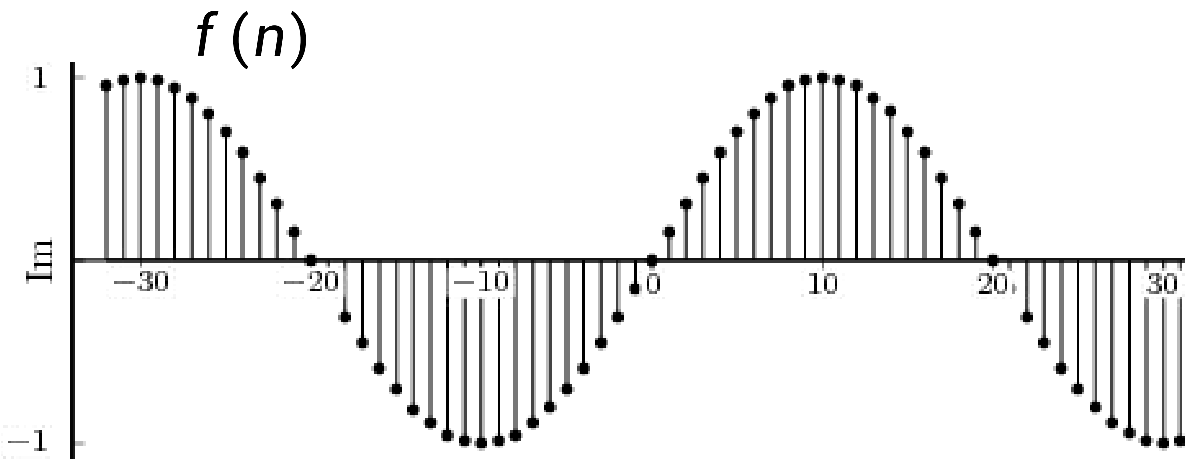
The number T is called **a period of** f.

If f is nonconstant (非定数), we define the **fundamental period (基本周期)**, or simply, the **period (周期)** of f to be **the smallest positive number** T for which (1.1) holds.

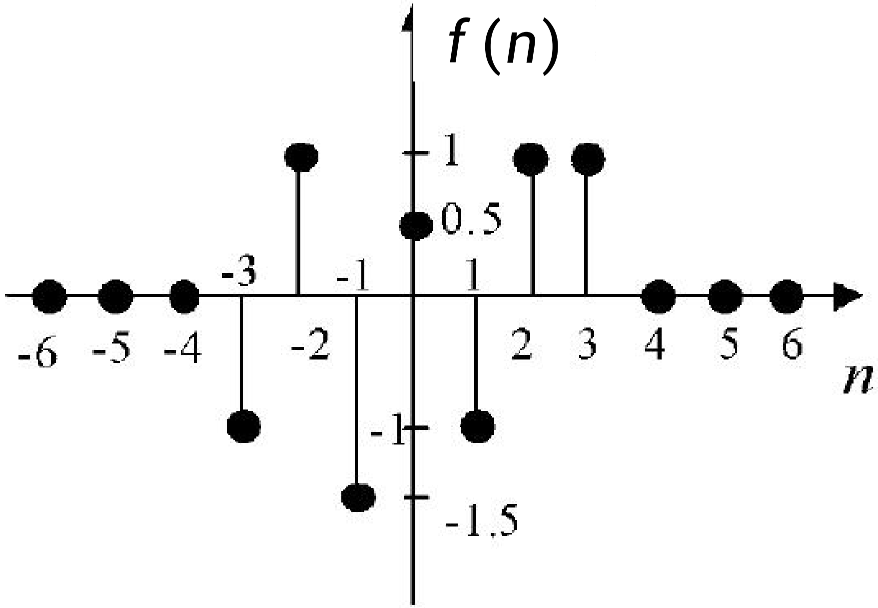
8.1 Discrete Signal 離散信号

Periodic Discrete-Time Signal

- A **discrete-time signal** $f(n)$ is called **periodic** with **fundamental period** $N \in \mathbf{Z}$ $(N > 0)$, if $f(n+N) = f(n)$ for all n.
- Otherwise it is called **aperiodic (非周期的).**



discrete-time periodic signal



discrete-time aperiodic signal

Figure 8.2

8.1 Discrete Signal 離散信号

Basic signals

- The **unit sample** or **unit impulse sequence** $\delta(n)$, defined as

$$\delta(n) = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}$$

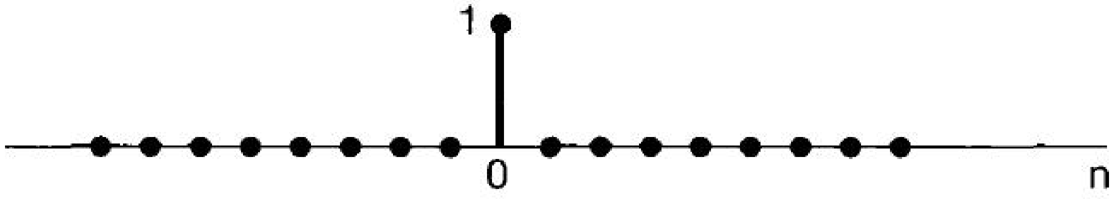


Figure 8.3

- The **unit step sequence** $u(n)$, defined as

$$u(n) = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

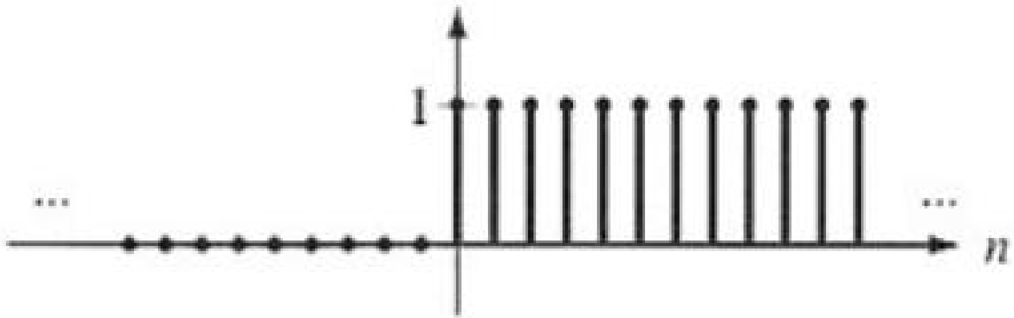


Figure 8.4

Unit impulse sequence is the **first difference of the discrete-time step sequence**. $\delta(n) = u(n) - u(n-1)$

Unit step sequence is the **running sum of the unit impulse sequence**. $u(n) = \sum_{k=-\infty}^{n} \delta(k)$

Sampling Property

The **unit impulse sequence** can be used to sample the value of a signal at $n = 0$.

In particular, since $\delta(n)$ is nonzero (and equal to 1) only for $n = 0$, it follows that

$$f(n)\delta(n) = f(0)\delta(n)$$

More generally, if we consider a **unit impulse** $\delta(n - n_0)$ at $n = n_0$, then

$$f(n)\delta(n - n_0) = f(n_0)\delta(n - n_0)$$

This is called **sampling property (サンプリング特性)**.

8.1 Discrete Signal 離散信号

Consider the signal $f\,(n)$

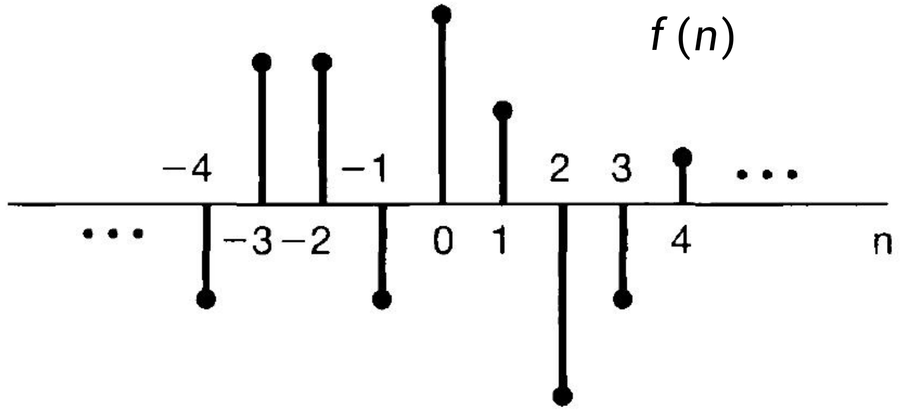


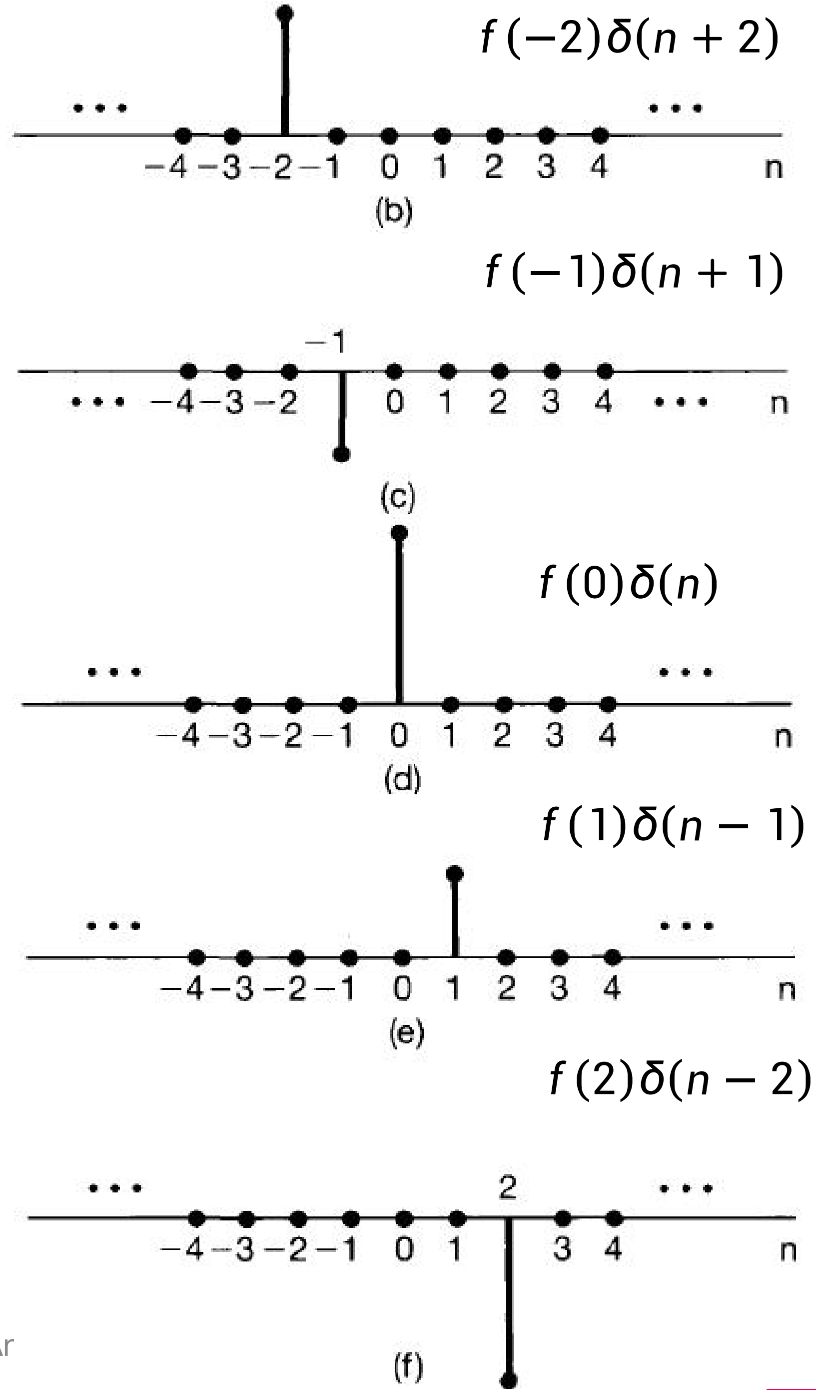
Figure 8.5 (a)

$$f\,(-1)\delta(n+1)=\begin{cases} f\,(-1), & n=-1 \\ 0, & n\neq -1 \end{cases}$$

$n_0=-1$

$$f\,(0)\delta(n)=\begin{cases} f\,(0), & n=0 \\ 0, & n\neq 0 \end{cases}$$

$$f\,(1)\delta(n-1)=\begin{cases} f\,(1), & n=1 \\ 0, & n\neq 1 \end{cases}$$



8.1 Discrete Signal 離散信号

Therefore, the sum of the five sequences in the Figure 8.5 (b)~(f) equals $f(n)$ for $-2 \leq n \leq 2$.

More generally, by including additional shifted, scaled impulses, we can write

$$f(n) = \ldots + f(-3)\delta(n+3) + f(-2)\delta(n+2) + f(-1)\delta(n+1) + f(0)\delta(n) + f(1)\delta(n-1) + f(2)\delta(n-2) + f(3)\delta(n-3) + \ldots \tag{8.1}$$

For any value of n, only one of the terms on the right-hand side of eq. (8.1) is nonzero, and the scaling associated with that term is precisely $f(n)$.

Writing this summation in a more compact form, we have

$$f(n) = \sum_{k=-\infty}^{+\infty} f(k)\delta(n-k) \tag{8.2}$$

This corresponds to **the representation of an arbitrary sequence as a linear combination of shifted unit impulses** $\delta(n-k)$, where **the weights in this linear combination are** $f(k)$.

*8.2 Sampling Theorem

サンプリング定理

* is the optional knowledge.

*8.2 Sampling Theorem サンプリング定理

Under certain conditions, a continuous-time signal can be completely represented by and recoverable from knowledge of its values, or samples, at points equally spaced in time.

In general, in the absence of any additional conditions or information, we would not expect that a signal could be uniquely specified by a sequence of equally spaced samples.

For example, in Figure 8.6 we illustrate **three different continuous-time signals**, all of which have identical values at integer multiples of T; that is, $f_1(kT) = f_2(kT) = f_3(kT)$

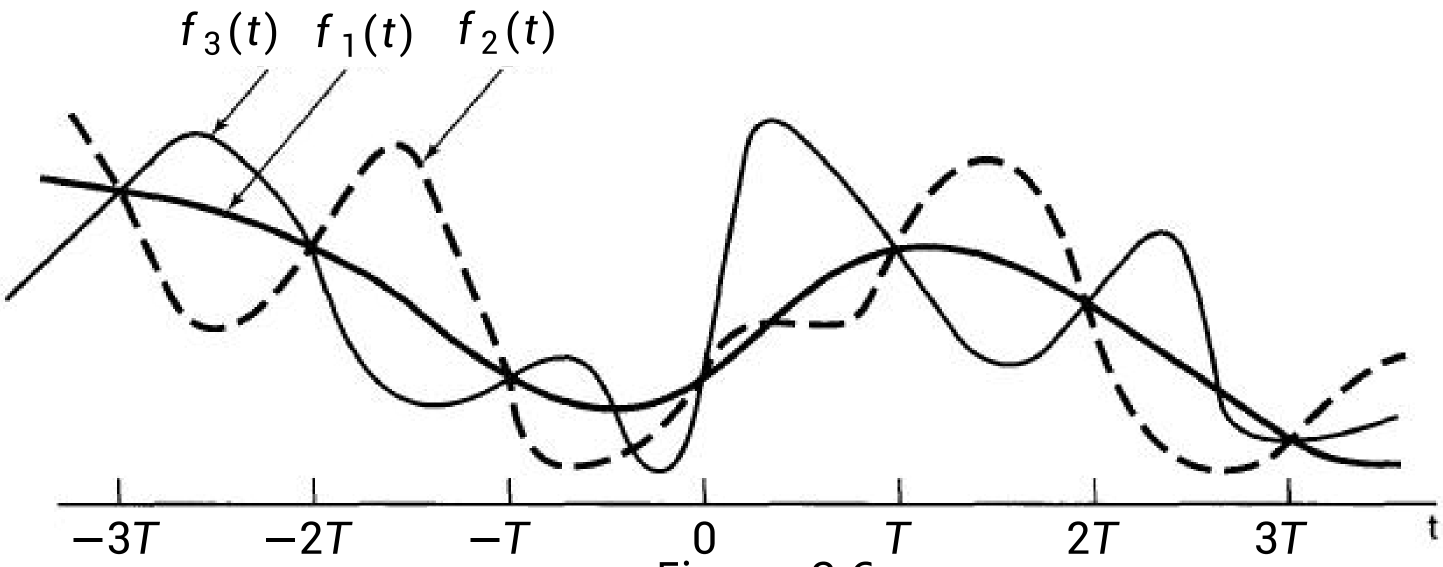


Figure 8.6

*8.2 Sampling Theorem サンプリング定理

In order to develop the **sampling theorem**, we need a convenient way in which to represent the sampling of a **continuous-time signal at regular intervals**.

A useful way to do this is **through the use of a periodic impulse train multiplied by the continuous-time signal** $f(t)$ that we wish to sample. This mechanism, known as **impulse-train sampling**, is depicted in Figure 8.7.

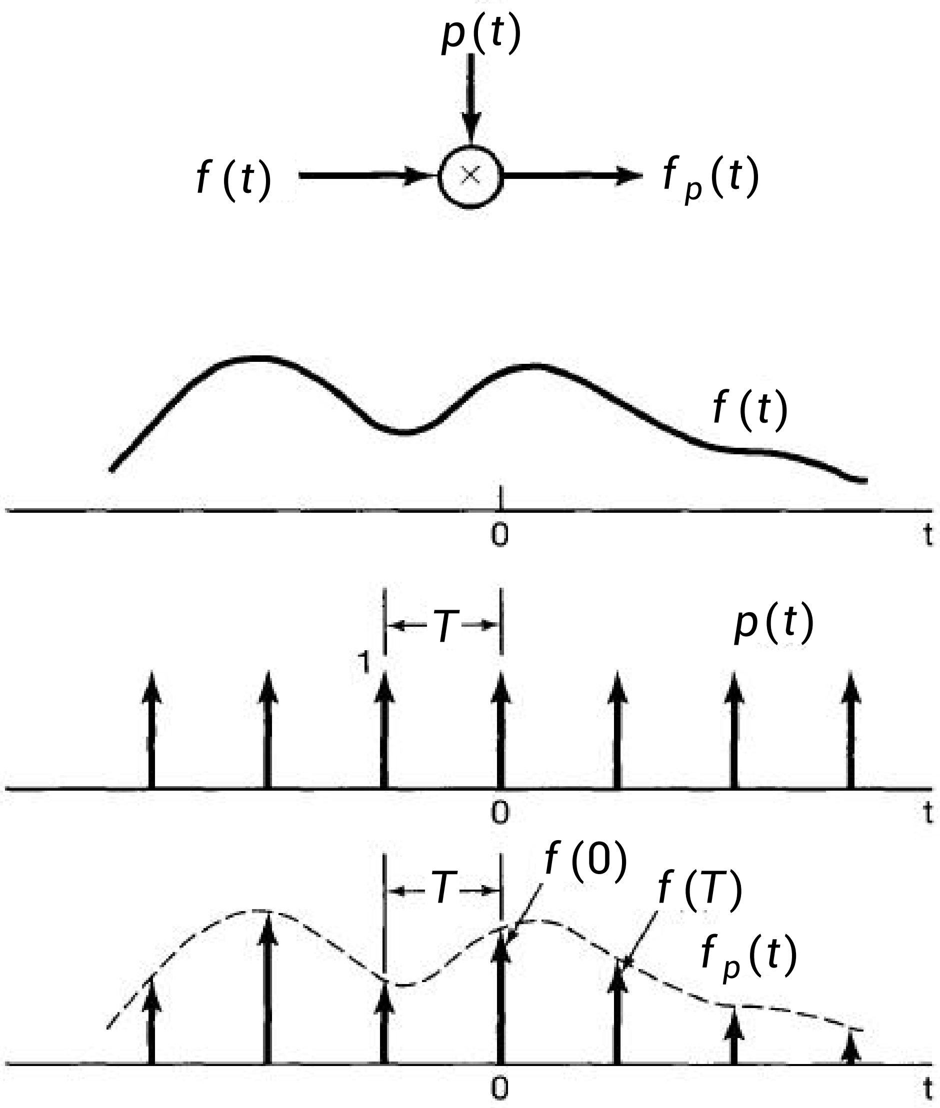


Figure 8.7

*8.2 Sampling Theorem サンプリング定理

This mechanism, known as **impulse-train sampling**.
The periodic impulse train $p(t)$ is referred to as the **sampling function**, the period T as the **sampling period**, and the **fundamental frequency** of $p(t)$, $\omega_s = \frac{2\pi}{T}$, as the **sampling frequency**.
In the **time domain**,

$$f_p(t) = f(t)p(t)$$

where

$$p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT) \tag{8.3}$$

Then

$$f_p(t) = \sum_{n=-\infty}^{+\infty} f(nT)\delta(t - nT) \tag{8.4}$$

Spectrum (スペクトル)

In analogy with the terminology used for the Fourier series coefficients of a periodic signal, the transform $\hat{f}(\omega)$ of an aperiodic signal $f(t)$ is commonly referred to as the **spectrum** of $f(t)$, as it provides us with the information needed for describing $f(t)$ as a linear combination (specifically, an integral) of sinusoidal signals at different frequencies.

*8.2 Sampling Theorem サンプリング定理

For (8.3) and (8.4), we have $\hat{p}(\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{+\infty}\delta(\omega - k\omega_s)$ and $\widehat{f_p}(\omega) = \frac{1}{T}\sum_{k=-\infty}^{+\infty}\hat{f}(\omega - k\omega_s)$

$\widehat{f_p}(\omega)$ is a **periodic function** of ω consisting of a superposition of shifted replicas of $\hat{f}(\omega)$, scaled by $1/T$, as illustrated in Figure 8.8.

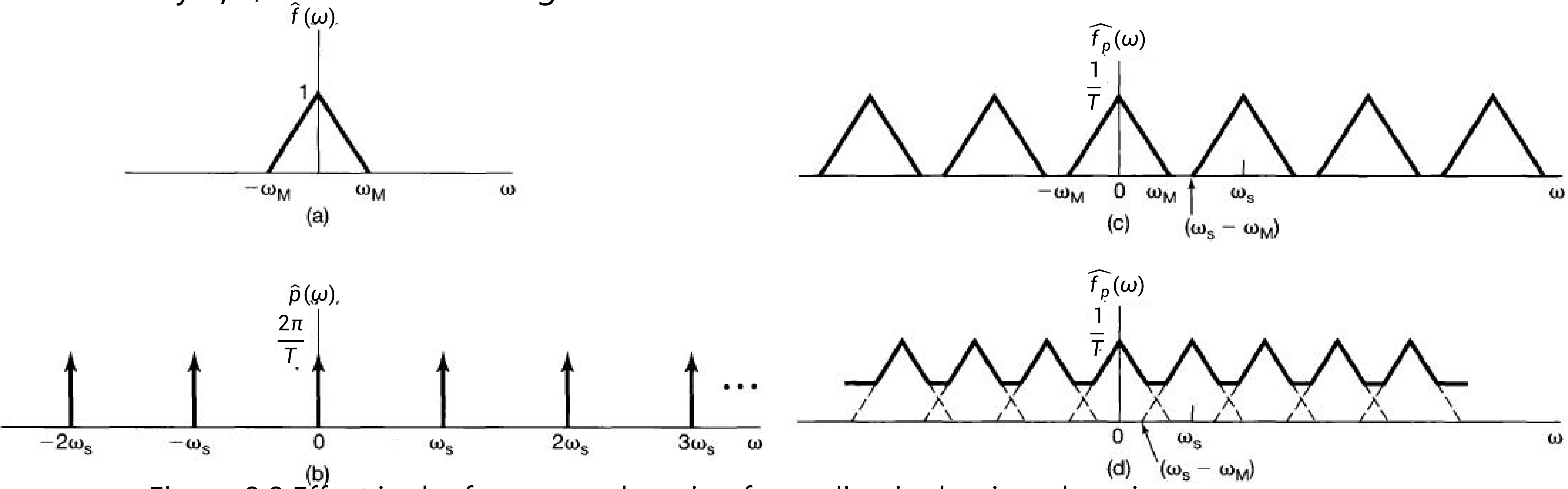


Figure 8.8 Effect in the frequency domain of sampling in the time domain:
(a) Spectrum of original signal; (b) Spectrum of sampling function;
(c) Spectrum of sampled signal with $\omega_s > 2\omega_M$; (d) Spectrum of sampled signal with $\omega_s < 2\omega_M$.

*8.2 Sampling Theorem サンプリング定理

Band Limited Function 帯域制限信号

A function $f(x)$ is called **band limited** if its **Fourier Transform $\hat{f}(\omega)$ vanishes outside a finite interval**. In this case, there is a positive number W such that $\hat{f}(\omega) = 0$ for all $|\omega| > W$. Any such number W is called a **band width (帯域幅)** of f.

- It is important to note that only $\hat{f}$ is required to vanish outside a finite interval and not f.
- Indeed, it can be shown that f and $\hat{f}$ cannot both vanish off a finite interval.
- Usually, the "most" functions are not band limited. We can show that at least all the functions that we have dealt with in this lecture can be approximated as closely as we want by band limited functions.

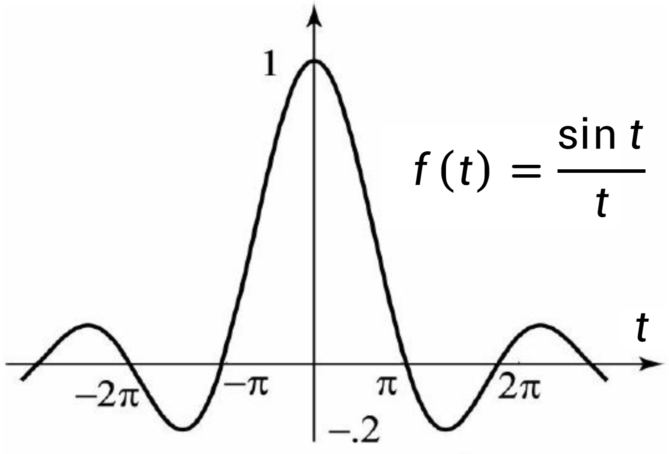
*8.2 Sampling Theorem サンプリング定理

Example 8.1 A band limited function

The function $f(t) = \frac{\sin t}{t}$ is shown in Figure 8.8. From the **Table of Fourier Transforms**, we have

$$\hat{f}(\omega) = \begin{cases} \sqrt{\frac{\pi}{2}} & if\ |\omega| < 1 \\ \frac{1}{2}\sqrt{\frac{\pi}{2}} & if\ \omega = \pm 1 \\ 0 & if\ |\omega| > 1 \end{cases}$$

Since $\hat{f}$ vanishes for all $|\omega| > 1$, we conclude that s is band limited with band width 1.



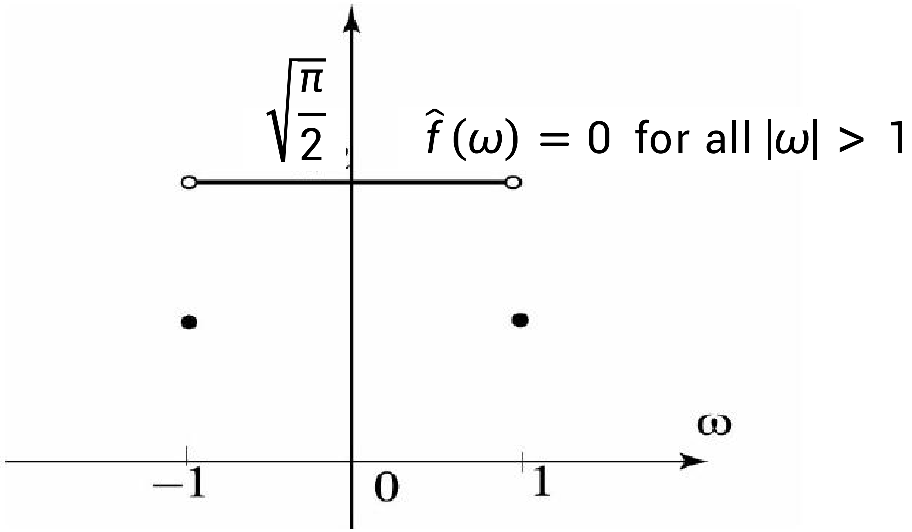


Figure 8.8 (a)The function $f(t) = \frac{\sin t}{t}$ is band limited with band widt (b)The Fourier transform of $f(t) = \frac{\sin t}{t}$ vanishes for all $|\omega| > 1$

*8.2 Sampling Theorem サンプリング定理

Theorem 8.1 Properties of Band Limited Functions

Suppose that a, b are **real constants** and $f(t)$, $g(t)$ are **functions.**

(a) If $f(t)$ and $g(t)$ **are band limited with band widths** W_1 and W_2, respectively, then $af + bg$ **is band limited with band width** W **smaller than or equal to the larger of** W_1 and W_2.

(b) If $f(t)$ **is band limited with band width** W, **then the translate of** $f(t)$ **by** a, $f(t-a)$, **is also band limited with band width** W.

(c) If either $f(t)$ or $g(t)$ **is band limited with band width** W, **then the convolution** $f * g$ **is also band limited with band width** W.

*8.2 Sampling Theorem サンプリング定理

Theorem 8.2 Sampling Theorem

Let $f(t)$ be a **band-limited signal** with $\hat{f}(\omega)= 0$ for $|\omega| > 2\omega_M$. Then $f(t)$ is uniquely determined by its samples $f(nT)$, $n = 0, \pm 1, \pm 2, \dots$, if

$$\omega_s > 2\omega_M \tag{8.5}$$

where

$$\omega_s > \frac{2\pi}{T}$$

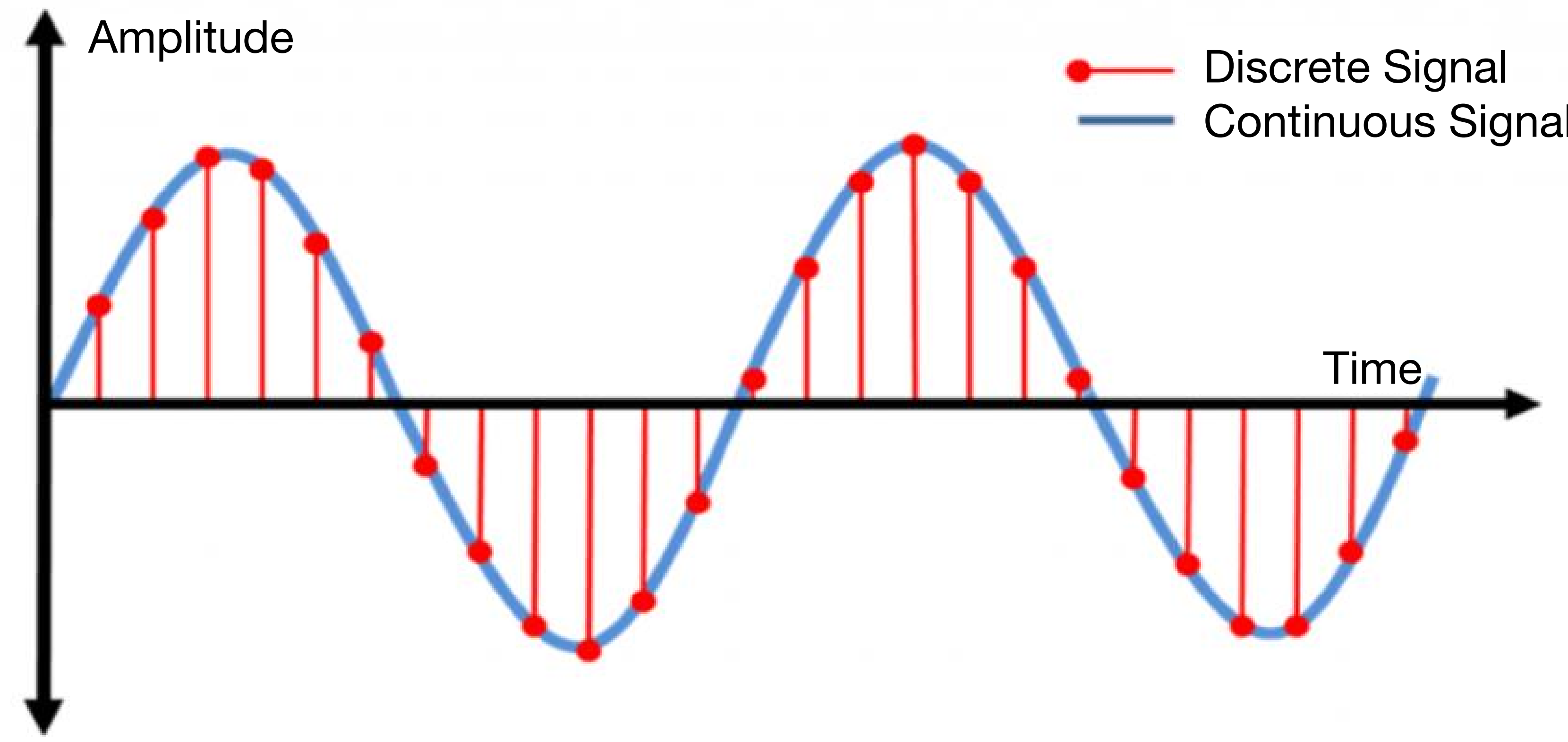
Given these samples, we can reconstruct $f(t)$ by generating a periodic impulse train in which successive impulses have amplitudes that are successive sample values.

This **impulse train** is then processed through an **ideal lowpass filter (理想的なローパスフィルター)** with gain T and **cutoff frequency (遮断周波数)** greater than ω_M and less than $\omega_s - \omega_M$. The resulting output signal will exactly equal $f(t)$.

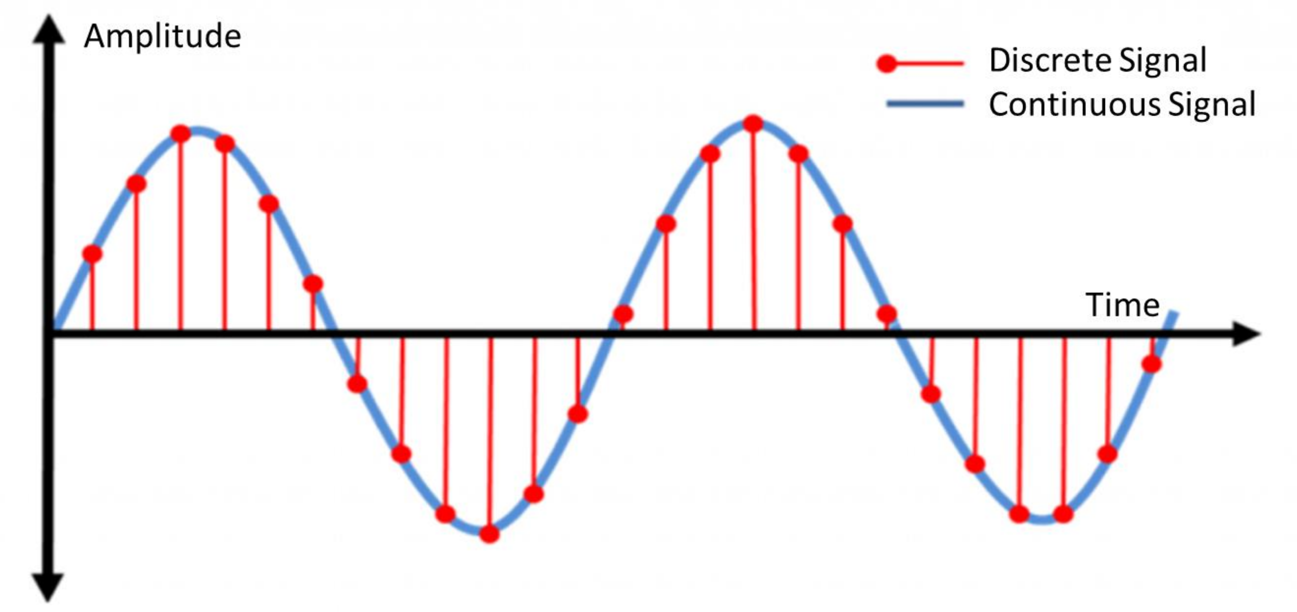
The **minimum sampling rate** of $\omega_s = 2\omega_M$ is called the **Nyquist rate**.

*8.2 Sampling Theorem サンプリング定理

The **sampling theorem** is a striking result that states that **certain functions can be reconstructed completely** from a discrete set of measurements or samples taken at equal intervals.



*8.2 Sampling Theorem サンプリング定理

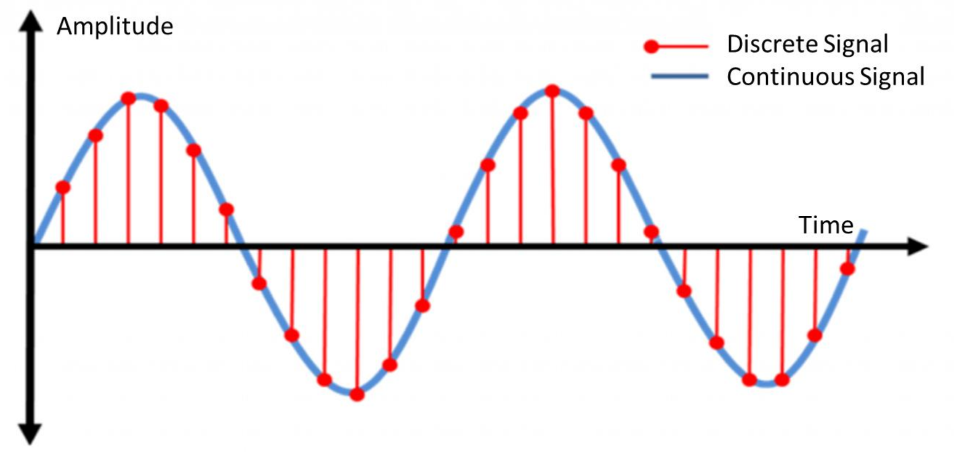


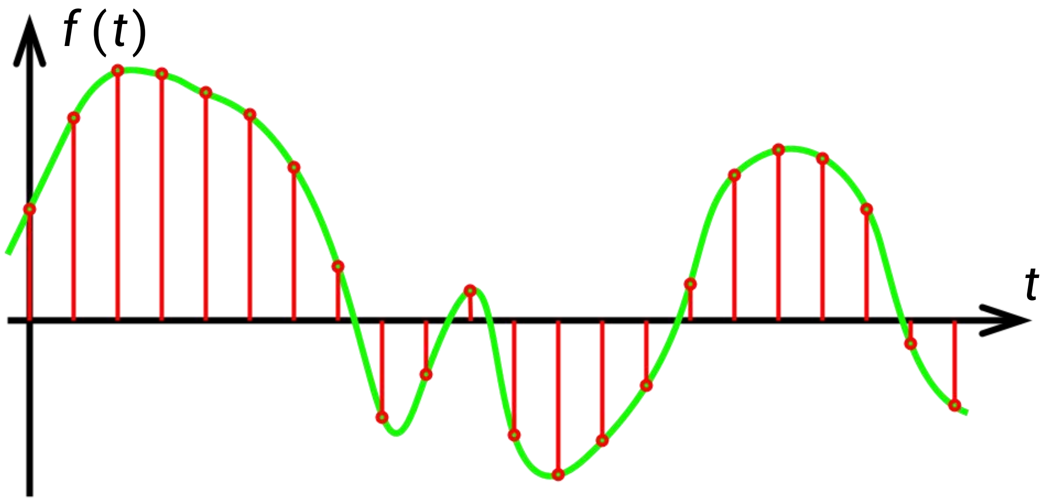
Let continuous function $f(t)$ be periodic, for simplicity of period 2π. We assume that N measurements of $f(t)$ are taken over the interval $0 \le t \le 2\pi$ at regularly spaced points

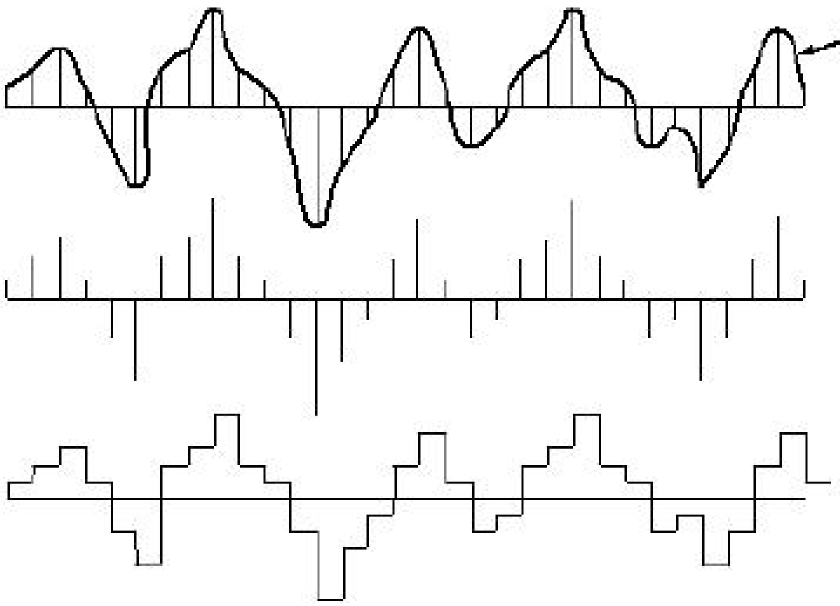
$$t_n = \frac{2\pi n}{N} \qquad\qquad n = 0, 1, \cdots, N-1$$

We also say that $f(t)$ is being **sampled** at these points.

*8.2 Sampling Theorem サンプリング定理







Sometimes, it can reconstruct the signal in this non-smooth way.

*8.2 Sampling Theorem サンプリング定理

Q: What if the Sampling Theorem is not satisfied?

Recall the signal in Figure 8.6.

In general, in the absence of any additional conditions or information, we would not expect that a signal could be uniquely specified by a sequence of equally spaced samples.

For example, in Figure 9.6 we illustrate **three different continuous-time signals**, all of which have identical values at integer multiples of T; that is, $f_1(kT) = f_2(kT) = f_3(kT)$

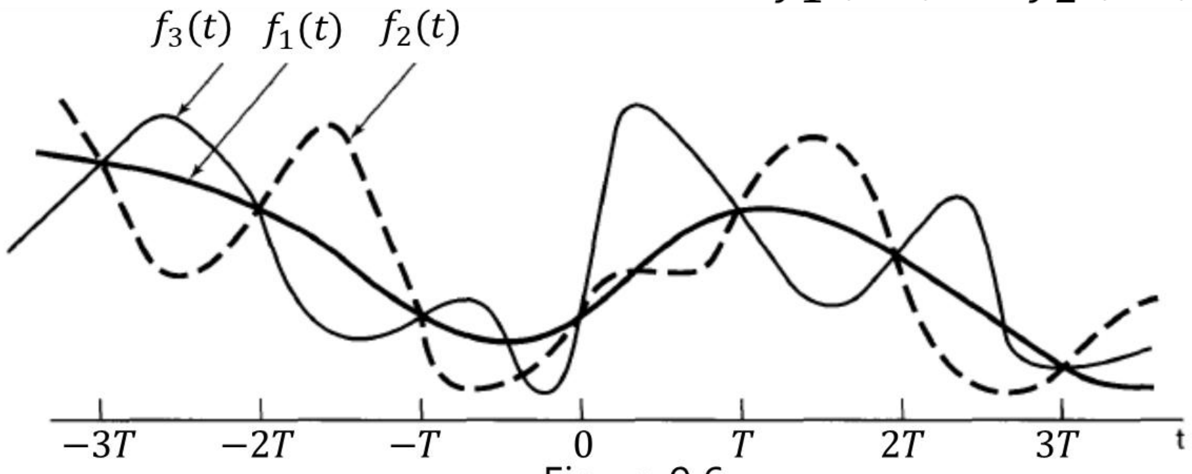


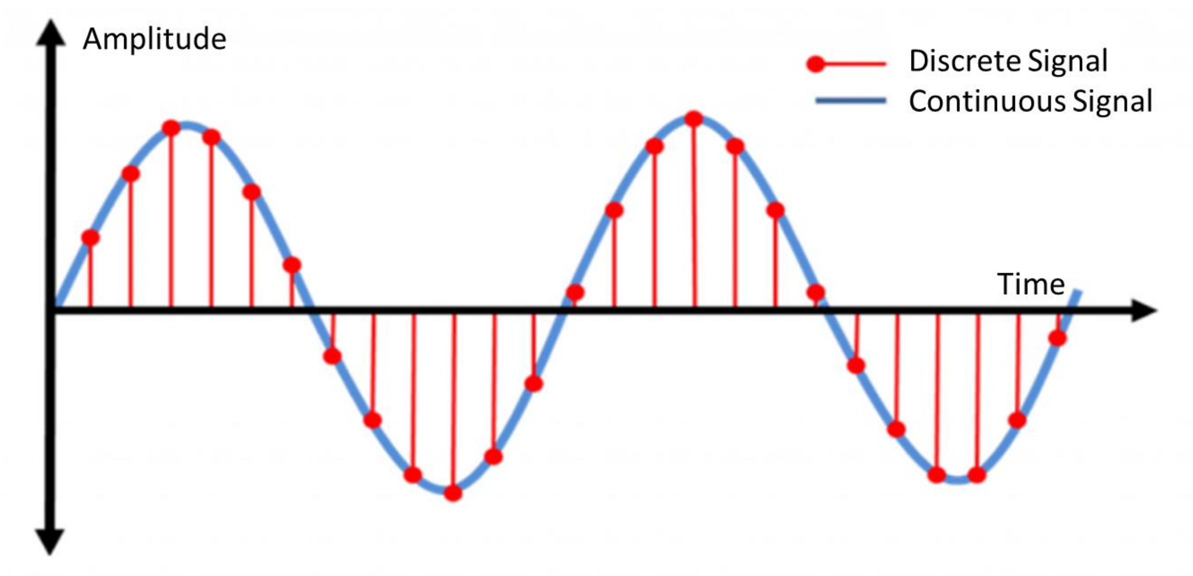
Figure 9.6

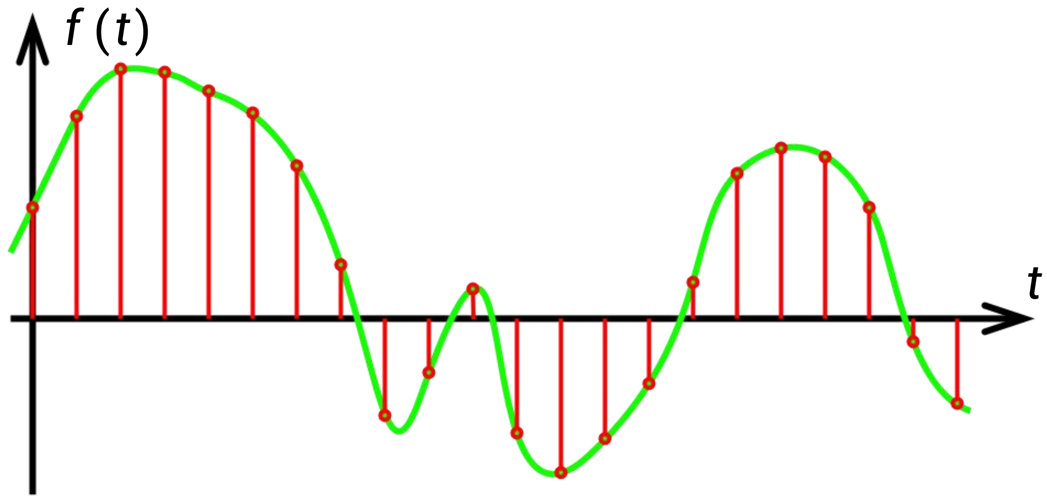
8.3 Discrete Fourier Transform

離散フーリエ変換

8.3 Discrete Fourier Transform 離散フーリエ変換

We are considering the discrete-time signals in this lecture.





8.3 Discrete Fourier Transform 離散フーリエ変換

The sequence shown below in Figure 8.10 is considered to be **one period of the periodic sequence in plot** (b).

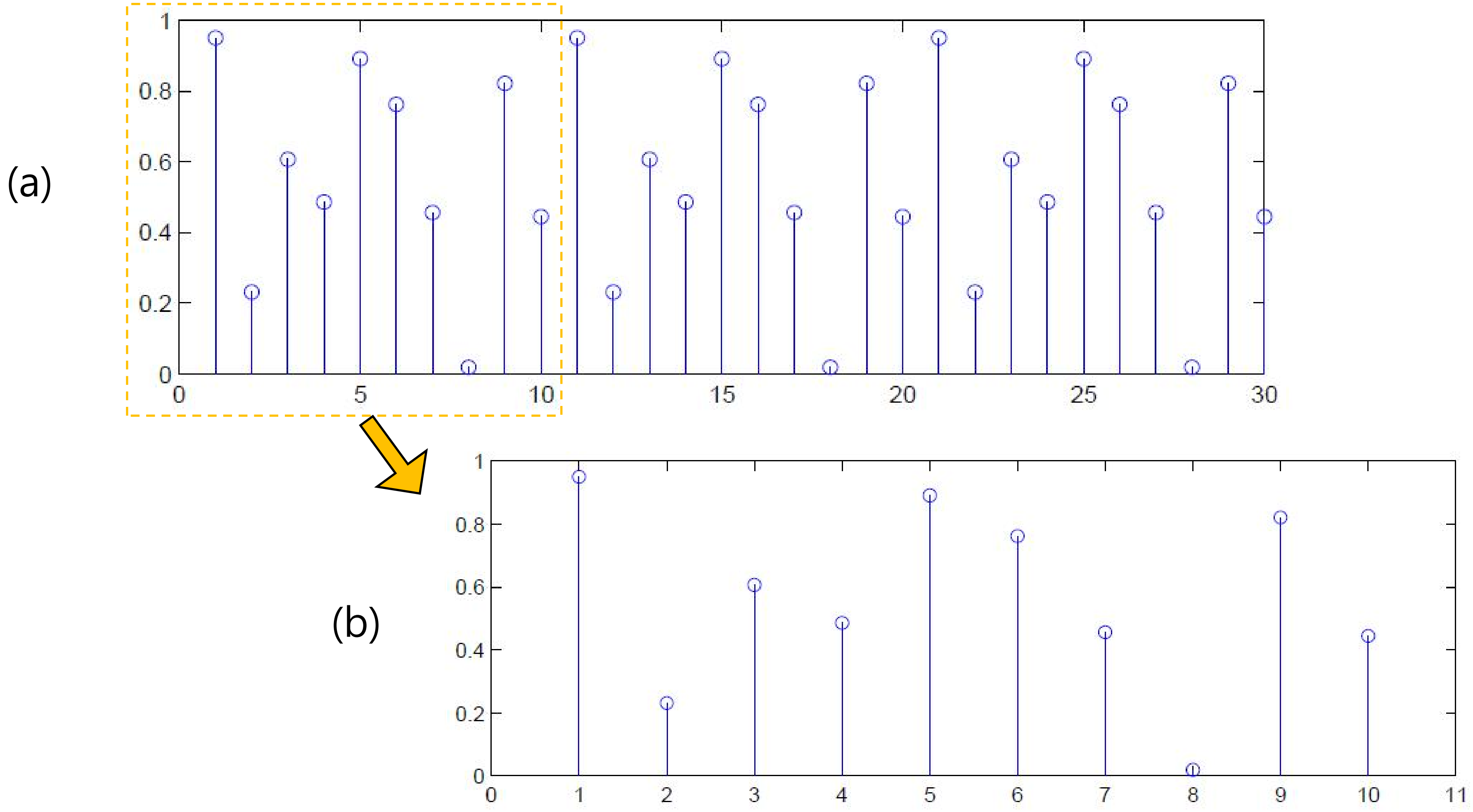
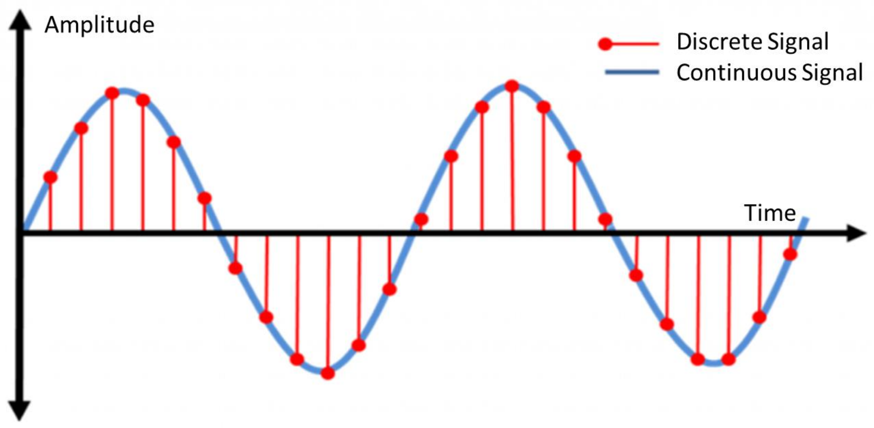


Figure 8.10: (a) Sequence of $N = 10$ samples. (b) Implicit periodicity.

8.3 Discrete Fourier Transform 離散フーリエ変換



Now very often a function is given only in terms of values at finitely many points, and one is interested in extending Fourier analysis to this case.

The main application of such a "Discrete Fourier Analysis" concerns large amounts of **equally spaced data**, as they occur in telecommunication, time series analysis, and various simulation problems.

In these situations, **dealing with sampled values rather than with functions**, we can replace the Fourier Transform by the so-called **Discrete Fourier Transform (DFT)**

8.3 Discrete Fourier Transform 離散フーリエ変換

Let these N samples be denoted as $f(n) = \{f(0), f(1), f(2), \cdots, f(k), \cdots, f(N-1)\}$.

We think of this as a function over the discrete domain $\{0,\ 1,\ \ldots,\ N-1\}$.

Basic Operations on Sequences

If f and g are two sequences, we define their sum $f+g$ and their product fg, by

$$(f+g)_k = f_k + g_k$$

respectively,

$$(fg)_k = f_k g_k$$

Also, if f is a sequence and a is a real number, then by af_k we mean the sequence whose kth term is af_k.

8.3 Discrete Fourier Transform 離散フーリエ変換

Discrete Fourier Transform

(Notice: The definitions of DFT and IDFT are consistent with wikipedia but not the textbook.)

The **Discrete Fourier Transform (DFT)** transforms a sequence of N real or complex numbers $f(n) = \{f(0), f(1), f(2), \cdots, f(N-1)\}$ into another sequence of complex numbers $\hat{f}(k)$ which is defined by

$$\hat{f}(k) = \sum_{n=0}^{N-1} f(n)\, e^{-i\frac{2\pi}{N}nk} \qquad k = 0,\ 1,\ \cdots,\ N-1 \qquad (8.6)$$

$$= \sum_{n=0}^{N-1} f(n) \left[\cos\frac{2\pi}{N}nk - i\sin\frac{2\pi}{N}nk\right]$$

where the last expression follows from the first one by Euler's formula.

The output is a complex number which encodes **the amplitude and phase** of a sinusoidal wave.

The Discrete Fourier Transform (DFT) is the equivalent of the continuous Fourier Transform for signals known only at N instants separated by sample times T (i.e. a finite sequence of data).

8.3 Discrete Fourier Transform 離散フーリエ変換

- Note the similarity with the definition of the Fourier transform. Here we multiply the values of the sequence by a complex exponential and then sum over the whole domain of the sequence.
- With the Fourier transform, we multiply the function by a complex exponential and integrate over the whole domain of the function.
- In fact, you will see that many properties of the Fourier transform will be translated to properties of the DFT by an appropriate conversion involving changing an integral over a continuous domain to a sum over a discrete domain.

8.3 Discrete Fourier Transform 離散フーリエ変換

The effect of computing the $\hat{f}(k)$ is to find the coefficients of an approximation of the signal by a linear combination of such waves.

Since each wave has an integer number of cycles per N time units, the approximation will be periodic with period N. This approximation is given by the **inverse Fourier transform.**

Like other transforms that we have encountered previously, the DFT has an inverse, known as the **Inverse Discrete Fourier Transform (IDFT),** given by

Inverse Discrete Fourier Transform

$$f(n) = \frac{1}{N}\sum_{k=0}^{N-1} \hat{f}(k)\, e^{i\frac{2\pi}{N}nk} \qquad n = 0,\ 1,\ \cdots,\ N-1 \qquad (8.7)$$

8.3 Discrete Fourier Transform 離散フーリエ変換

Therefore, we have the DFT and IDFT pair:

Discrete Fourier Transform (DFT)

$$\hat{f}(k) = \sum_{n=0}^{N-1} f(n)\, e^{-i\frac{2\pi}{N}nk} \qquad k = 0,\ 1,\ \cdots,\ N-1 \qquad (8.6)$$

Inverse Discrete Fourier Transform (IDFT)

$$f(n) = \frac{1}{N}\sum_{k=0}^{N-1} \hat{f}(k)\, e^{i\frac{2\pi}{N}nk} \qquad n = 0,\ 1,\ \cdots,\ N-1 \qquad (8.7)$$

8.3 Discrete Fourier Transform 離散フーリエ変換

The DFT is useful in many applications, including the simple signal spectral analysis outlined above. Knowing how a signal can be expressed as a combination of waves allows for manipulation of that signal and comparisons of different signals:

•**Digital files (jpg, mp3, etc.)** can be shrunk by eliminating contributions from the least important waves in the combination.

•**Different sound files** can be compared by comparing the coefficients $\hat{f}(k)$ of the DFT.

•**Radio waves** can be filtered to avoid "noise" and listen to the important components of the signal.

8.3 Discrete Fourier Transform 離散フーリエ変換

Let $f(n) = \{f(0), f(1), f(2), \cdots, f(N-1)\} = \{1,\ 0,\ 0,\ \cdots,\ \ 0,\ \cdots,\ 0\}$

Then the DFT of the $f(n)$ is $\quad \hat{f}(k) = \sum_{n=0}^{N-1} f(n)\, e^{-i\frac{2\pi}{N}nk} = 1 \cdot e^{-i\frac{2\pi}{N}\cdot 0\cdot k} + 0 \cdot e^{-i\frac{2\pi}{N}\cdot 1\cdot k} + \cdots + 0 \cdot e^{-i\frac{2\pi}{N}\cdot (N-1)\cdot k} + \ldots = 1$

$k = 0,1,2,\ldots,\ N-1$

8.3 Discrete Fourier Transform 離散フーリエ変換

Example 8.2 Find the Discrete Fourier Transform (DFT) of the sequence $\{f(0), f(1), f(2), f(3)\} = \{0,\ 1,\ 0,\ 0\},\ N = 4.$

Solution:

By (8.6), we have $\hat{f}(k) = \sum_{n=0}^{3} f(n)\, e^{-i\frac{2\pi}{4}nk}$

$$\hat{f}(0) = f(0)e^{-i\frac{2\pi}{4}\cdot 0\cdot 0} + f(1)e^{-i\frac{2\pi}{4}\cdot 1\cdot 0} + f(2)e^{-i\frac{2\pi}{4}\cdot 2\cdot 0} + f(3)e^{-i\frac{2\pi}{4}\cdot 3\cdot 0}$$

$$= 0\cdot e^{-i\frac{2\pi}{4}\cdot 0\cdot 0} + 1\cdot e^{-i\frac{2\pi}{4}\cdot 1\cdot 0} + 0\cdot e^{-i\frac{2\pi}{4}\cdot 2\cdot 0} + 0\cdot e^{-i\frac{2\pi}{4}\cdot 3\cdot 0}$$

$$= e^0 \quad = 1$$

$$\hat{f}(1) = f(0)e^{-i\frac{2\pi}{4}\cdot 0\cdot 1} + f(1)e^{-i\frac{2\pi}{4}\cdot 1\cdot 1} + f(2)e^{-i\frac{2\pi}{4}\cdot 2\cdot 1} + f(3)e^{-i\frac{2\pi}{4}\cdot 3\cdot 1}$$

$$= 0\cdot e^{-i\frac{2\pi}{4}\cdot 0\cdot 1} + 1\cdot e^{-i\frac{2\pi}{4}\cdot 1\cdot 1} + 0\cdot e^{-i\frac{2\pi}{4}\cdot 2\cdot 1} + 0\cdot e^{-i\frac{2\pi}{4}\cdot 3\cdot 1}$$

$$\underbrace{= e^{-i\frac{\pi}{2}} \quad = \cos\frac{\pi}{2} - i\sin\frac{\pi}{2}}_{\text{By Euler's Formula}} \quad = -i$$

8.3 Discrete Fourier Transform 離散フーリエ変換

Solution (cont.):

$$\hat{f}(2) = f(0)e^{-i\frac{2\pi}{4}\cdot 0\cdot 2} + f(1)e^{-i\frac{2\pi}{4}\cdot 1\cdot 2} + f(2)e^{-i\frac{2\pi}{4}\cdot 2\cdot 2} + f(3)e^{-i\frac{2\pi}{4}\cdot 3\cdot 2}$$

$$= 0\cdot e^{-i\frac{2\pi}{4}\cdot 0\cdot 2} + 1\cdot e^{-i\frac{2\pi}{4}\cdot 1\cdot 2} + 0\cdot e^{-i\frac{2\pi}{4}\cdot 2\cdot 2} + 0\cdot e^{-i\frac{2\pi}{4}\cdot 3\cdot 2}$$

$$= e^{-i\pi} \quad = (\cos\pi - i\sin\pi) \quad = -1$$

$$\hat{f}(3) = f(0)e^{-i\frac{2\pi}{4}\cdot 0\cdot 3} + f(1)e^{-i\frac{2\pi}{4}\cdot 1\cdot 3} + f(2)e^{-i\frac{2\pi}{4}\cdot 2\cdot 3} + f(3)e^{-i\frac{2\pi}{4}\cdot 3\cdot 3}$$

$$= 0\cdot e^{-i\frac{2\pi}{4}\cdot 0\cdot 3} + 1\cdot e^{-i\frac{2\pi}{4}\cdot 1\cdot 3} + 0\cdot e^{-i\frac{2\pi}{4}\cdot 2\cdot 3} + 0\cdot e^{-i\frac{2\pi}{4}\cdot 3\cdot 3}$$

$$= e^{-i\frac{3\pi}{2}} \quad = \left(\cos\frac{3\pi}{2} - i\sin\frac{3\pi}{2}\right) \quad = i$$

Thus $\{\hat{f}(0), \hat{f}(1), \hat{f}(2), \hat{f}(3)\} = \{1, -i, -1, i\},\ N = 4$

8.3 Discrete Fourier Transform 離散フーリエ変換

Example 8.3 Find the Discrete Fourier Transform (DFT) of the

sequence $\{f(0), f(1), f(2), f(3)\} = \{1,\ 2-i,\ -i,\ -1+2i\},\ N = 4.$

Solution:

By (8.6), we have $\hat{f}(k) = \sum_{n=0}^{3} f(n)\, e^{-i\frac{2\pi}{4}nk}$

$$\hat{f}(0) = f(0)e^{-i\frac{2\pi}{4}\cdot 0\cdot 0} + f(1)e^{-i\frac{2\pi}{4}\cdot 1\cdot 0} + f(2)e^{-i\frac{2\pi}{4}\cdot 2\cdot 0} + f(3)e^{-i\frac{2\pi}{4}\cdot 3\cdot 0}$$

$$= 1\cdot e^{-i\frac{2\pi}{4}\cdot 0\cdot 0} + (2-i)\cdot e^{-i\frac{2\pi}{4}\cdot 1\cdot 0} + (-i)\cdot e^{-i\frac{2\pi}{4}\cdot 2\cdot 0} + (-1+2i)\cdot e^{-i\frac{2\pi}{4}\cdot 3\cdot 0}$$

$$= 1 + (2-i) + (-i) + (-1+2i) \quad = 2$$

$$\hat{f}(1) = f(0)e^{-i\frac{2\pi}{4}\cdot 0\cdot 1} + f(1)e^{-i\frac{2\pi}{4}\cdot 1\cdot 1} + f(2)e^{-i\frac{2\pi}{4}\cdot 2\cdot 1} + f(3)e^{-i\frac{2\pi}{4}\cdot 3\cdot 1}$$

$$= 1\cdot e^{-i\frac{2\pi}{4}\cdot 0\cdot 1} + (2-i)\cdot e^{-i\frac{2\pi}{4}\cdot 1\cdot 1} + (-i)\cdot e^{-i\frac{2\pi}{4}\cdot 2\cdot 1} + (-1+2i)\cdot e^{-i\frac{2\pi}{4}\cdot 3\cdot 1}$$

$$= 1 + (2-i)\cdot e^{-i\frac{\pi}{2}} + (-i)\cdot e^{-i\pi} + (-1+2i)\cdot e^{-i\frac{3\pi}{2}} \quad = -2-2i$$

8.3 Discrete Fourier Transform 離散フーリエ変換

Solution (cont.):

$$\hat{f}(2) = f(0)e^{-i\frac{2\pi}{4}\cdot 0\cdot 2} + f(1)e^{-i\frac{2\pi}{4}\cdot 1\cdot 2} + f(2)e^{-i\frac{2\pi}{4}\cdot 2\cdot 2} + f(3)e^{-i\frac{2\pi}{4}\cdot 3\cdot 2}$$

$$= 1\cdot e^{-i\frac{2\pi}{4}\cdot 0\cdot 2} + (2-i)\cdot e^{-i\frac{2\pi}{4}\cdot 1\cdot 2} + (-i)\cdot e^{-i\frac{2\pi}{4}\cdot 2\cdot 2} + (-1+2i)\cdot e^{-i\frac{2\pi}{4}\cdot 3\cdot 2}$$

$$= 1 + (2-i)\cdot e^{-i\pi} + (-i)\cdot e^{-i2\pi} + (-1+2i)\cdot e^{-i3\pi} \quad = -2i$$

$$\hat{f}(3) = f(0)e^{-i\frac{2\pi}{4}\cdot 0\cdot 3} + f(1)e^{-i\frac{2\pi}{4}\cdot 1\cdot 3} + f(2)e^{-i\frac{2\pi}{4}\cdot 2\cdot 3} + f(3)e^{-i\frac{2\pi}{4}\cdot 3\cdot 3}$$

$$= 1\cdot e^{-i\frac{2\pi}{4}\cdot 0\cdot 3} + (2-i)\cdot e^{-i\frac{2\pi}{4}\cdot 1\cdot 3} + (-i)\cdot e^{-i\frac{2\pi}{4}\cdot 2\cdot 3} + (-1+2i)\cdot e^{-i\frac{2\pi}{4}\cdot 3\cdot 3}$$

$$= 1 + (2-i)\cdot e^{-i\frac{3\pi}{2}} + (-i)\cdot e^{-i3\pi} + (-1+2i)\cdot e^{-i\frac{9\pi}{2}} \quad = 4+4i$$

Thus $\{\hat{f}(0), \hat{f}(1), \hat{f}(2), \hat{f}(3)\} = \{2, -2-2i, -2i, 4+4i\},\ N = 4$

Example 8.4 Find the Inverse Discrete Fourier Transform (IDFT) of the sequence $\{\hat{f}(0), \hat{f}(1), \hat{f}(2), \hat{f}(3)\} = \{1, -i, -1, i\}$, $N = 4$.

Solution: By (8.7), we have

$$f(n) = \frac{1}{N}\sum_{k=0}^{3} \hat{f}(k)\, e^{i\frac{2\pi}{N}nk}$$

$$f(0) = \frac{1}{4}\left[\hat{f}(0)e^{i\frac{2\pi}{4}\cdot 0\cdot 0} + \hat{f}(1)e^{i\frac{2\pi}{4}\cdot 0\cdot 1} + \hat{f}(2)e^{i\frac{2\pi}{4}\cdot 0\cdot 2} + \hat{f}(3)e^{i\frac{2\pi}{4}\cdot 0\cdot 3}\right]$$

$$= 0$$

$$f(1) = \frac{1}{4}\left[\hat{f}(0)e^{i\frac{2\pi}{4}\cdot 1\cdot 0} + \hat{f}(1)e^{i\frac{2\pi}{4}\cdot 1\cdot 1} + \hat{f}(2)e^{i\frac{2\pi}{4}\cdot 1\cdot 2} + \hat{f}(3)e^{i\frac{2\pi}{4}\cdot 1\cdot 3}\right]$$

$$= \frac{1}{4}\left[1\cdot 1 + (-i)\cdot e^{i\frac{\pi}{2}} + (-1)\cdot e^{i\pi} + i\cdot e^{i\frac{3\pi}{2}}\right]$$

$$= \frac{1}{4}[1 + (-i)\cdot i + (-1)\cdot(-1) + i\cdot(-i)]$$ (By Euler's formula $e^{i\theta} = \cos\theta + i\sin\theta$)

$$= 1$$

8.3 Discrete Fourier Transform 離散フーリエ変換

Solution (cont.):

$$f(2) = \frac{1}{4}\left[\hat{f}(0)e^{i\frac{2\pi}{4}\cdot 2\cdot 0} + \hat{f}(1)e^{i\frac{2\pi}{4}\cdot 2\cdot 1} + \hat{f}(2)e^{i\frac{2\pi}{4}\cdot 2\cdot 2} + \hat{f}(3)e^{i\frac{2\pi}{4}\cdot 2\cdot 3}\right]$$

$$= \frac{1}{4}[1\cdot 1 + (-i)\cdot e^{i\pi} + (-1)\cdot e^{i2\pi} + i\cdot e^{i3\pi}]$$

$$= \frac{1}{4}[1 + (-i)\cdot(-1) + (-1)\cdot 1 + i\cdot(-1)]$$

$$= 0$$

$$f(3) = \frac{1}{4}\left[\hat{f}(0)e^{i\frac{2\pi}{4}\cdot 3\cdot 0} + \hat{f}(1)e^{i\frac{2\pi}{4}\cdot 3\cdot 1} + \hat{f}(2)e^{i\frac{2\pi}{4}\cdot 3\cdot 2} + \hat{f}(3)e^{i\frac{2\pi}{4}\cdot 3\cdot 3}\right]$$

$$= \frac{1}{4}\left[1\cdot 1 + (-i)\cdot e^{i\frac{3\pi}{2}} + (-1)\cdot e^{i3\pi} + i\cdot e^{i\frac{9\pi}{2}}\right]$$

$$= \frac{1}{4}[1 + (-i)\cdot(-i) + (-1)\cdot(-1) + i\cdot i]$$

$$= 0$$

Thus $\{f(0), f(1), f(2), f(3)\} = \{0,1,0,0\},\ N = 4$

8.3 Discrete Fourier Transform 離散フーリエ変換

Theorem 8.3 Transforms of Convolutions

For two N-sequences f and g we have

$$\mathcal{F}_N(f * g) = \mathcal{F}_N(f)\mathcal{F}_N(g)$$

$$\mathcal{F}_N^{-1}(\mathcal{F}_N(f)\mathcal{F}_N(g)) = f * g$$

Review for Lecture 8

- Discrete Signal
- Sampling Theorem
- Discrete Fourier Transform (DFT) and Inverse Discrete Fourier Transform (IDFT)
- Convolution

Assignment

Please Check https://web-ext.u-aizu.ac.jp/~xiangli/teaching/MA05/index.html

Reading materials: Lecture 8 Slides

References

[1] Nakhlé H. Asmar, *Partial Differential Equations with Fourier Series and Boundary Value Problems 2nd Edition*, 2004

[2] Alan V. Oppenheim, Alan S. Willsky and S. Hamid, *Signals and Systems 2nd Edition*, Pearson, 1997.

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